Paradoxes and Problems in the Causal Interpretation of Equilibrium Economics

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New and Old Ideas in Macroeconomics

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Today's discussion

1. Motivation
2. Process and equilibrium models
3. Equilibrium and mechanisms
4. Temporal non-locality
5. Immaculate revelation

Motivation (1)

In the case of the equilibrium account of wage and employment determination, parameters describing the degree of intertemporal substitutability do the job (in an empirical sense) of the parameter describing auctioneer behavior in the Phillips curve model. On these parameters, we have a wealth of inexpensively available data from census cohort information, from panel data describing the reactions of individual households to a variety of changing market conditions, and so forth. In principle (and perhaps before too long, in practice, for there is a good deal of very promising research going on on just this topic[13]) these crucial parameters can be estimated independently from individual as well as aggregate data. If so, we will know what the aggregate parameters mean, we will understand them in a sense that disequilibrium adjustment parameters will *never* be understood. This is exactly why we care about the "microeconomic foundations" of aggregate theories.[14]

Lucas (1980)

Motivation (2)

It is the task of comparative statics to show the determination of the equilibrium values of given variables (unknowns) under postulated conditions (functional relationships) with various data (parameters) being specified. Thus, in the simplest case of a partial-equilibrium market for a single commodity, the two independent relations of supply and demand, each drawn up with other prices and institutional data being taken as given, determine by their intersection the equilibrium quantities of the unknown price and quantity sold. If no more than this could be said, the economist would be truly vulnerable to the gibe that he is only a parrot taught to say "supply and demand." Simply to know that there are efficacious "laws" determining equilibrium tells us nothing of the character of these laws. In order for the analysis to be useful it must provide information concerning the way in which our equilibrium quantities will change as a result of changes in the parameters taken as independent data.

Samuelson (1941)

Overview

Generalized Interdependence is the crown jewel of economics but:

- Equilibrium assumptions distort our understanding of the models that we write
- They cannot be endowed with "causal" stories: give rise to a series of paradox that cannot make sense in any process model

Relevant to the core questions in economics: whether flexible prices ensure full employment or efficiently aggregate dispersed information.

$\hookrightarrow$Our equilibrium requirements assume away these problems by collapsing them into solution concepts.

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What is a model?

A set of agents $\mathcal{A}$, endogenous variables $y = (y_i)_{i \in I} \in Y$, exogenous variables $z = (z_j)_{j \in J} \in Z$, and a collection of conditions:

$$y_{L_k} \in \Gamma_k(x_{R_k}) \qquad k \in K \qquad x = (y, z) \in X = Y \times Z$$

Here $L_k \subseteq I$ indexes the outputs of condition k, and $R_k \subseteq I \cup J$ its inputs; $\Gamma_k : X_{R_k} \rightrightarrows Y_{L_k}$ may be a correspondence.

Its reduced form is the set of outcomes consistent with all conditions:

$$Y^*(z) = \{y \in Y : \ y_{L_k} \in \Gamma_k(x_{R_k}) \text{ for every } k \in K\}$$

$\hookrightarrow$ An equation may describe behaviour, an institutional rule, book-keeping, or a physical process. Its economic interpretation is part of the model, it cannot be read off its mathematical form.

The four principles of a process model

1. Recursiveness and time's arrow. The graph of dependencies is acyclic, and inputs precede outputs in time.

2. Non-empty outcomes. Each correspondence Γ_k has at least one output for every admissible input.

3. One determination per variable. Every endogenous variable appears as the output of exactly one condition.

4. Who does what? Unless determined by rules, identities, or physical processes, a variable is chosen by an agent. A behavioural condition contains only that agent's choices as outputs, and only what she knows or experiences before choosing as inputs.

$\hookrightarrow$ A process model describes how the outcome is generated. An equilibrium model, in Dogra's terminology, contains behavioural conditions and violates at least one of these principles.

The distinction is not in the reduced form

Consider a reduced-form VAR and one of its structural representations:

$$y_t = B_1 y_{t-1} + u_t \qquad \Longleftrightarrow \qquad A_0 y_t = A_1 y_{t-1} + \varepsilon_t$$

$$B_1 = A_0^{-1} A_1 \qquad u_t = A_0^{-1} \varepsilon_t$$

The same observed dynamics can have different structural interpretations.

- Contemporaneous links may summarize a sequence of decisions occurring within the observation period: simultaneity may arise from time aggregation.
- They may instead impose equilibrium relations between decisions without specifying a process making these relations hold.

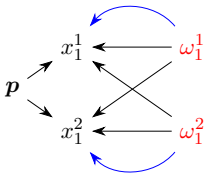
$\hookrightarrow$ Calling a system "structural", or solving it recursively on paper, does not settle the issue. We must say what each condition means and which agent determines each variable.

Example 1: a Walrasian exchange economy

m consumers, ℓ goods; endowments ω_i, preferences u_i.
Consumption x_i and prices p are endogenous.

$$x_i \in \arg\max_{x \in \mathbb{R}^{\ell}_{++}} u_i(x) \quad \text{s.t. } p \cdot (x - \omega_i) \leq 0$$

$$\sum_{i=1}^{m} (x_i^j - \omega_i^j) = 0 \qquad j = 1, \ldots, \ell$$



(a) Competitive equilibrium

- Demands are constrained both by optimality and by market clearing: Principle 3 fails.
- No agent chooses prices; no rule describes their determination: Principle 4 fails.

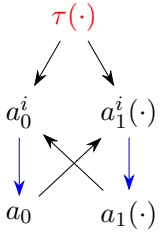
↪ Market clearing restricts the outcome. It does not describe how prices adjust or how incompatible demands are handled.

Example 2: rational expectations

Two dates, a continuum $i \in [0,1]$, and $\theta \sim \mathcal{N}(0,1)$ drawn at date 1.
Policy is a function $\tau(\theta)$ committed to at date 0.

$$a_0^i = \alpha \mathbb{E}_0 \tau(\theta) + \beta \mathbb{E}_0 a_1(\theta)$$

$$a_1^i(\theta) = \theta + \tau(\theta) + \delta a_0 \qquad a_t = \int_0^1 a_t^i \, di$$



(c) REE

Date 1 endogenous variables are functions of the state.

For $1 - \beta\delta \neq 0$, the reduced form implies

$$a_0 = \frac{\alpha + \beta}{1 - \beta\delta} \mathbb{E}\tau(\theta)$$

$\hookrightarrow$ Principle 1 fails: RE assume a fixed-point of the mapping that takes expectation of endogenous variables to their realized values $\rightarrow$ Joint distribution of outcomes and expectations

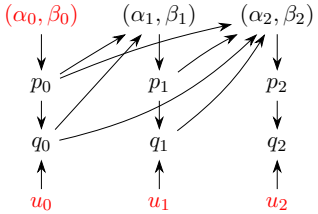
Example 3: learning about demand

A firm believes demand is $\alpha_t - \beta_t p$ and faces production cost $q^2/2$:

$$p_t \in \arg\max_{p \geq 0} \left\{ p(\alpha_t - \beta_t p) - \tfrac{1}{2}(\alpha_t - \beta_t p)^2 \right\}$$

Actual demand is $q_t = a - bp_t + u_t$, with u_t white noise. The firm then updates its estimates:

$$(\alpha_{t+1}, \beta_{t+1}) \in \arg\min_{\alpha \geq 0} \; \min_{\beta \geq \underline{\beta} > 0} \sum_{\tau=0}^{t} (q_\tau - \alpha + \beta p_\tau)^2$$



(d) Learning about demand

$\hookrightarrow$ Past data $\rightarrow$ beliefs $\rightarrow$ chosen price $\rightarrow$ observed demand $\rightarrow$ revised beliefs.

The firm optimizes, but takes as given beliefs determined before it acts. Each variable has a place in an explicit process.

Optimization is not the dividing line

Let $f_i = \partial f / \partial x_i$ denote marginal products and d_i commodity demands. Compare two superficially identical systems:

$$x \in \{x \geq 0 : \ f_i(x) = w_i \quad i = 1, \ldots, \ell\}$$
$$p \in \{p \geq 0 : \ d_i(p) = \omega_i \quad i = 1, \ldots, \ell\}$$

- In the first, a firm jointly chooses factor inputs to maximize profits given prices. With a well-defined optimum, this can be a process block.
- In the second, market clearing jointly restricts prices. No individual is described as choosing them.

Block-recursive does not mean that every choice within one person's decision problem must be causally ordered.

$\hookrightarrow$ But calling the *whole economy* a single block would make recursiveness vacuous. Principle 4 requires an account of individual action.

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What does it mean to understand a model?

A general equilibrium model imposes simultaneous restrictions:

$$\mathcal{H}(x, y, z) = 0 \qquad \Longleftrightarrow \qquad f_k(x, y, z) = 0 \quad \text{for all } k$$

To describe one endogenous variable's causal effect on another, we need a directional relation such as

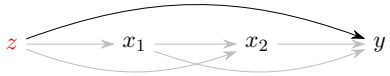
$$y_i = f_i(y_j, \ldots) \qquad \text{where "=" is interpreted as "is caused by"}$$

In a process model, causal ordering gives this interpretation to relations between blocks. An equilibrium condition alone does not select a direction.

$\hookrightarrow$ Understanding should help us anticipate what changes when we perturb parameters, environments, or policies, without solving every perturbed model.

Direct and indirect effects

In a recursive causal model, an intervention can fix the mediators x and vary z, or retain only a particular path from z to y.



(a) Direct effect of z on y



(b) Indirect effect via pathway $z \to x_1 \to y$

- Direct effect: vary z while holding the mediators fixed.
- Indirect effects: the remaining effect, through the mediators; in a linear system it can be added up over the individual paths.

$\hookrightarrow$ The causal structure tells us which relations an intervention replaces. In a simultaneous equilibrium system, that interpretation is not supplied by the equations alone.

A simple example: Werning (2022)

Start from the New Keynesian Phillips curve:

$$\pi_t = \kappa x_t + \beta \mathbb{E}_t \pi_{t+1}$$

Does expected inflation "cause" current inflation, with coefficient β? Under the terminal condition $\lim_{K \to \infty} \beta^{K+1} \mathbb{E}_t \pi_{t+K+1} = 0$, the same equation gives

$$\pi_t = \kappa \mathbb{E}_t \sum_{k=0}^{\infty} \beta^k x_{t+k}$$

Now only current and expected output gaps appear. Or we could write

$$x_t = \kappa^{-1} \left(\pi_t - \beta \mathbb{E}_t \pi_{t+1} \right)$$

$\hookrightarrow$ These are equivalent restrictions on equilibrium paths. None tells us, by itself, which variable is determined by which

The RANK example: the household

Perfect foresight. The representative household takes real interest rates, wages, and profit transfers $\{r_t, w_t, T_t\}_{t\geq 0}$ as given and solves

$$\max_{\{c_t, n_t, b_{t+1}\}} \sum_{t=0}^{\infty} \beta^t \left[\frac{c_t^{1-\gamma}}{1-\gamma} - \phi \frac{n_t^{1+\nu}}{1+\nu} \right]$$

$$c_t + \frac{b_{t+1}}{1+r_t} = w_t n_t + T_t + b_t \qquad b_0 = 0$$

with a no-Ponzi condition. Here γ^{-1} is the IES; ν^{-1} is the Frisch elasticity and $\phi > 0$ scales labour disutility.

$$c_t^{-\gamma} = \beta(1+r_t) c_{t+1}^{-\gamma} \qquad w_t c_t^{-\gamma} = \phi n_t^{\nu}$$

$\hookrightarrow$ This provides a consumption function $c_0 = C_0(\{r_t, w_t, T_t\}_{t\geq 0})$ But the variables the household takes as given are endogenous in GE.

The RANK example: equilibrium

Monopolistic firms, linear production and Rotemberg price adjustment costs; a Taylor rule and a balanced government budget. The log-linearized system is

$$y_t = y_{t+1} - \gamma^{-1}(i_t - \pi_{t+1}) \qquad \pi_t = \kappa y_t + \beta \pi_{t+1} \qquad i_t = \varphi \pi_t + \varepsilon_t$$

Here y_t is output's log-deviation from steady state; $\kappa > 0$ and $\varphi > 1$. Consider a one-off shock, $\varepsilon_0 \neq 0$ and $\varepsilon_t = 0$ for $t \geq 1$.

$$y_t = \pi_t = i_t = 0 \quad (t \geq 1)$$

$$y_0 = -\frac{\varepsilon_0}{\varphi\kappa + \gamma} \qquad \pi_0 = -\frac{\kappa\varepsilon_0}{\varphi\kappa + \gamma} \qquad i_0 = \frac{\gamma\varepsilon_0}{\varphi\kappa + \gamma}$$

$\hookrightarrow$ The unique bounded solution returns to steady state from date 1. At date 0, output, inflation and the interest rate are determined jointly.

The Kaplan–Moll–Violante decomposition

Return to the household's consumption function and totally differentiate:

$$dc_0 = \underbrace{\sum_{t=0}^{\infty} \frac{\partial C_0}{\partial r_t}\, dr_t}_{\text{direct effect}} + \underbrace{\sum_{t=0}^{\infty} \frac{\partial C_0}{\partial w_t}\, dw_t + \sum_{t=0}^{\infty} \frac{\partial C_0}{\partial T_t}\, dT_t}_{\text{indirect effects}}$$

- Direct: change interest rates, holding the paths of wages and profit transfers fixed.
- Indirect: the consumption response associated with changes in wages and transfers along the equilibrium path.

The familiar story: lower rates encourage consumption, greater demand raises income, higher income encourages additional consumption.

$\hookrightarrow$ This is a valid chain-rule calculation. Does it also identify a causal sequence within the equilibrium model?

Deriving the direct effect

Hold $w_t = \bar{w}$ and $T_t = \bar{T}$ fixed; perturb only r_0. Differentials are in levels. Let c_+ and n_+ denote the constant choices from date 1 onwards. Euler and labour supply imply

$$c_+ = [\beta(1+r_0)]^{1/\gamma} c_0 \qquad n_s = (\bar{w}/\phi)^{1/\nu} c_s^{-\gamma/\nu} \quad s \in \{0,+\}$$

The lifetime budget is

$$c_0 + \frac{c_+}{(1+r_0)(1-\beta)} = \bar{w} n_0 + \bar{T} + \frac{\bar{w} n_+ + \bar{T}}{(1+r_0)(1-\beta)}$$

At steady state, $\bar{c} = \bar{w}\bar{n} + \bar{T}$ and $1 + \bar{r} = \beta^{-1}$. Discount-factor terms cancel, writing $\ell = \gamma \bar{w} \bar{n} / (\nu \bar{c})$ gives

$$(1+\ell)\left(dc_0 + \frac{\beta}{1-\beta} dc_+\right) = 0 \Longrightarrow dc_+ = dc_0 + \frac{\beta \bar{c}}{\gamma} dr_0$$

$$\hookrightarrow \quad dc_0^D = -\frac{\beta^2 \bar{c}}{\gamma}\, dr_0$$

Almost all intertemporal substitution?

In general equilibrium, the one-off shock leaves c_1 unchanged. The Euler equation therefore gives the total response:

$$dc_0 = -\frac{\beta\bar{c}}{\gamma} dr_0 = -\frac{\beta\bar{c}}{\gamma} \left(\underbrace{\beta}_{\text{direct share}} + \underbrace{1-\beta}_{\text{indirect share}} \right) dr_0$$

For β close to one, almost all of the consumption response is "accounted for" by the direct effect.

Shouldn't the IES then be a key determinant of the effect of monetary policy?

$\hookrightarrow$ The decomposition concerns the response along a particular equilibrium path. It does not yet tell us how that path changes when the parameters themselves change.

Counterexample 1: a real-rate rule

Replace the Taylor rule by $i_t = r_t + \varphi\pi_t + \varepsilon_t$. Together with the Fisher equation $i_t = r_t + \pi_{t+1}$,

$$\pi_{t+1} = \varphi\pi_t + \varepsilon_t$$

For the one-off shock and $\varphi > 1$, boundedness and the Phillips curve imply

$$\pi_0 = -\frac{\varepsilon_0}{\varphi} \qquad y_0 = -\frac{\varepsilon_0}{\kappa\varphi} \qquad r_0 = i_0 = \frac{\gamma}{\kappa\varphi}\varepsilon_0$$

The direct share is still $\beta \approx 1$. Yet, holding κ and φ fixed, the output response is independent of γ^{-1}.

$\hookrightarrow$ The other equations pin down output. The IES determines the real rate consistent with that output response.

In the background: the weird "off-equilibrium" assumption of a Taylor rule: the CB stands ready to blow-up the economy if its chosen path is not respected.

Counterexample 2: working capital

Keep the standard Taylor rule, but change the supply side:

- Decreasing returns, $Y_t = n_t^{\alpha}$, with $0 < \alpha < 1$.
- Rigid real wages; firms borrow their wage bill at nominal rate i_t.

Financing costs now enter marginal cost, and hence the Phillips curve:

$$\pi_t = \kappa y_t + q\, i_t + \beta \pi_{t+1} \qquad q := \frac{\kappa\alpha}{1-\alpha}$$

For a one-off shock, $i_0 = -\gamma y_0$. Substitution into the Taylor rule gives

$$-\gamma y_0 = \varphi(\kappa - q\gamma) y_0 + \varepsilon_0 \quad \Longrightarrow \quad y_0 = -\frac{\varepsilon_0}{\varphi\kappa + \gamma(1-\varphi q)}$$

$\hookrightarrow$ The direct share remains β, but the IES now interacts with the financing-cost effect on inflation.

The IES can even enter with the opposite sign

Write $K = \varphi q = \varphi\kappa\alpha/(1-\alpha)$ and $D = \varphi\kappa + \gamma(1-K)$. Consider parameters with a unique bounded equilibrium and $D > 0$.

- If $K = 1$,
$$y_0 = -\frac{\varepsilon_0}{\varphi\kappa}: \quad \text{the IES drops out completely}$$
- If $K > 1$, a higher IES reduces the magnitude of the output response. With $s = \gamma^{-1}$ and other coefficients fixed,
$$\frac{\partial}{\partial s}\left|\frac{\partial y_0}{\partial \varepsilon_0}\right| = \frac{1-K}{s^2 D^2} < 0$$

$\hookrightarrow$ An almost entirely "intertemporal substitution" decomposition is consistent with an irrelevant IES, or with less stimulus from a higher IES.

Which equation tells the story?

In the real-rate-rule example, one could begin with the household:

$$y_0 = -\gamma^{-1} r_0 \qquad \rightsquigarrow \qquad \text{“real rates determine consumption”}$$

Or one could begin with monetary policy and the Phillips curve:

$$\pi_0 = -\varepsilon_0/\varphi \qquad y_0 = \pi_0/\kappa \qquad \rightsquigarrow \qquad \text{“inflation determines output”}$$

Both calculations give the same solution. They give different accounts of how the solution is reached.

$\hookrightarrow$ The household equation has no privileged role in determining aggregate consumption. All conditions must hold jointly.

Starting from another equation changes what appears as a channel or an intermediate variable. The system itself does not choose between these stories.

Proposition 3.1: the general linear system

Let y be the outcome, $x \in \mathbb{R}^{n-1}$ the other endogenous variables, and z an exogenous scalar:

$$Ax + by = cz \qquad y = g^\top x$$

The last equation describes an individual's choice of y given x. Define $M = A + bg^\top$. If M is invertible,

$$x = M^{-1}cz \qquad y = g^\top M^{-1}cz$$

Write $v = \partial x/\partial z = M^{-1}c$ and $Q = \partial y/\partial z = g^\top v \neq 0$. The decomposition share and the signed parameter elasticity are

$$s_i := g_i \frac{dx_i/dz}{dy_i/dz} \qquad \varepsilon_i = \frac{\partial Q}{\partial g_i}\frac{g_i}{Q} \qquad \sum_i s_i = 1$$

$\hookrightarrow$ Does a large share s_i imply that g_i matters strongly for Q?

Proposition 3.1: what feedback changes

Hold A, b, c and the other elements of g fixed. Since $Mv = c$,

$$M\frac{\partial v}{\partial g_i} + bv_i = 0 \qquad \frac{\partial v}{\partial g_i} = -M^{-1}bv_i$$

Differentiate $Q = g^\top v$:

$$\frac{\partial Q}{\partial g_i} = v_i + g^\top \frac{\partial v}{\partial g_i} = (1 - g^\top M^{-1}b)v_i$$

Hence

$$\epsilon_i = Fs_i \qquad F := 1 - g^\top M^{-1}b$$

- Recursive system: $b = 0$, A lower triangular with unit diagonal. Thus $F = 1$ and $\varepsilon_i = s_i$.
- Unrestricted system: changing g_i also changes the mediators' response. F may be zero or negative; shares alone do not identify elasticities.

Does the behavioural equation determine y?

An extreme case in Proposition 3.1:

$$b \notin \text{col}(A)$$

There is then a vector ℓ with $\ell^\top A = 0$ but $\ell^\top b \neq 0$. Multiplying the first block by $\ell^\top$ gives

$$\ell^\top b\, y = \ell^\top c\, z \qquad \Longrightarrow \qquad y = \frac{\ell^\top c}{\ell^\top b}\, z$$

As long as the full system remains uniquely solvable, changing g_i leaves the response of y unchanged: $\epsilon_i = 0$, whatever its decomposition share.

$\hookrightarrow$ This is what happens in the counterexamples. The household's parameters determine the values of other variables needed to make its optimal choice consistent with the outcome.

Relevance: consumption taxes versus interest rates

A temporary consumption-tax cut has the same effect on the household's intertemporal incentive as a lower interest rate. Let $\tau_0 = \Delta \log(1 + \text{tax}_0) - \Delta \log(1 + \text{tax}_1)$:

$$y_0 = -\gamma^{-1}(i_0 + \tau_0) \qquad \pi_1 = y_1 = 0$$

Use the working-capital example with $K = \varphi q = 1$, rigid real wages, and tax revenue rebated lump sum. The supply and policy equations still imply

$$(1 - \varphi q) i_0 = \varphi \kappa y_0 + \varepsilon_0 \quad \Longrightarrow \quad y_0 = -\frac{\varepsilon_0}{\varphi \kappa}$$

Thus a tax cut $\tau_0 < 0$, holding ε_0 fixed, leaves output unchanged:

$$\frac{\partial y_0}{\partial \tau_0} = 0 \qquad i_0 = -\gamma y_0 - \tau_0 \quad \Longrightarrow \quad di_0 = -d\tau_0$$

$\hookrightarrow$ The endogenous rise in interest rates offsets the tax incentive. Matching a partial-equilibrium channel need not match a policy's GE effect.

Proposition 3.2: substitute interventions

Add an intervention $\widetilde{z}$ that acts directly on the behavioural equation:

$$Ax + by = cz \qquad y = g^\top x + h\widetilde{z} \qquad h = g_i \frac{\partial x_i}{\partial z} = g_i v_i$$

It mimics the original treatment's effect through mediator x_i, holding x fixed. With $M = A + bg^\top$ invertible,

$$\frac{\partial x}{\partial \widetilde{z}} = -M^{-1}bh \qquad \frac{\partial y}{\partial \widetilde{z}} = h + g^\top \frac{\partial x}{\partial \widetilde{z}} = (1 - g^\top M^{-1} b)h$$

For $Q = \partial y / \partial z \neq 0$,

$$\frac{\partial y / \partial \widetilde{z}}{\partial y / \partial z} = F s_i$$

$\hookrightarrow$ Recursive: the ratio is s_i. Unrestricted: it may have the opposite sign, or be zero when $b \notin \mathrm{col}(A)$, whatever the size of s_i.

Can a process justify the equilibrium?

The decompositions are algebraically valid, but the usual causal stories can mislead us about:

- which parameters matter
- what happens when a channel is weakened in another environment
- whether a substitute intervention will work

There is a form of "incomensurability" between various equilibrium models.

$\hookrightarrow$ Perhaps a plausible adjustment process lies behind the equilibrium conditions: the spirit of Samuelson's correspondence principle.

But then we need to describe that process and establish the connection. The following examples show why its existence cannot simply be presumed.

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The ij game

At date t, a continuum of atomistic agents i chooses $x_i \in \mathbb{R}$. At $t+1$, a separate continuum j chooses $y_j \in \mathbb{R}$. Let $x = \int x_i \, di$, $y = \int y_j \, dj$, and

$$u_i(x_i; x, y) = yx_i \qquad u_j(y_j; x, y) = 0$$

The j agents are always indifferent. The i agents' best responses are

$$\arg\max_{x_i \in \mathbb{R}} yx_i = \begin{cases} \mathbb{R} & y = 0 \\ \varnothing & y \neq 0 \end{cases}$$

- In an equilibrium of the full game, necessarily $y = 0$.
- In the date $t+1$ subgame, after any history of x_i, every profile of y_j is optimal: any y is possible.

$\hookrightarrow$ How does assuming that the i agents optimized yesterday restrict the indifferent j agents' choices today?

The pricing game: ingredients and timing

A continuum of atomistic yeoman farmers $i \in [0,1]$, each producing one variety. Dates $t = 0, \ldots, T$. Preferences are

$$\sum_{t=0}^{T} \beta^t \left[\frac{c_t(i)^{1-\sigma}}{1-\sigma} - v(y_t(i)) \right] \qquad v', v'' > 0 \quad \sigma \geq 0$$

$$c_t(i) = \left[\int_0^1 c_t(i,j)^{(\gamma-1)/\gamma} \, dj \right]^{\gamma/(\gamma-1)} \qquad \gamma > 1$$

Here γ is the substitution elasticity between goods.

1. Morning: each farmer posts a freely chosen nominal price $p_t(i)$.
2. Afternoon: farmers observe prices and order goods, sellers must produce enough to meet demand. Purchases are made on credit.

$\hookrightarrow$ Prices are posted before purchases, but fully flexible. There is no auctioneer clearing markets before trade but sellers must meet demand.

The pricing game: demand and credit

Optimal allocation of spending across goods implies

$$c_t(i,j) = c_t(i)\left(\frac{p_t(j)}{p_t}\right)^{-\gamma} \qquad p_t = \left[\int_0^1 p_t(j)^{1-\gamma}\, dj\right]^{1/(1-\gamma)}$$

Writing $y_t = \int_0^1 c_t(i)\, di$, each farmer faces demand

$$y_t(i) = y_t\left(\frac{p_t(i)}{p_t}\right)^{-\gamma}$$

Credit has no limit and earns zero nominal interest. Balances evolve as

$$m_{t+1}(i) = m_t(i) + p_t(i)y_t(i) - p_t c_t(i) \qquad m_0(i) = 0$$

A sufficiently large terminal bankruptcy penalty induces $m_{T+1}(i) \geq 0$.

$\hookrightarrow$ These are explicit trading and settlement rules. The additional assumption is equilibrium behaviour within this game.

The full-game equilibrium

With perfect foresight, optimal pricing and consumption require

$$c_t(i)^{-\sigma}\frac{p_t(i)}{p_t} = \frac{\gamma}{\gamma-1}v'\left(y_t\left(\frac{p_t(i)}{p_t}\right)^{-\gamma}\right)$$

$$c_t(i)^{-\sigma} = \beta\frac{p_t}{p_{t+1}}c_{t+1}(i)^{-\sigma} \qquad (t < T)$$

Together with lifetime budgets and aggregation, these imply symmetry (Proposition 4.1): $p_t(i) = p_t$, $c_t(i) = y_t(i) = y_t$, and $m_{t+1}(i) = 0$.

$$1 = \frac{\gamma}{\gamma-1}(y^*)^{\sigma}v'(y^*) \qquad y_t = y^* \qquad p_{t+1} = \beta p_t$$

$\hookrightarrow$ The price level $p_0 > 0$ is indeterminate, but real output is always at "full employment" y^*. The monopoly markup makes this level inefficient.

The final afternoon: any level of activity?

Start in the afternoon of date T, with exactly the state reached in a full-game equilibrium:

$$p_T(i) = p_T \qquad m_T(i) = 0 \quad \text{for all } i$$

Each farmer's income is $p_T y_T$. Taking aggregate demand as given, she chooses

$$\max_{c_T(i) \geq 0} \left\{ \frac{c_T(i)^{1-\sigma}}{1-\sigma} - v(y_T) \right\} \quad \text{s.t. } p_T c_T(i) \leq p_T y_T$$

The constraint binds: $c_T(i) = y_T$. Aggregation gives

$$y_T = \int_0^1 c_T(i)\, di = y_T \quad \Longrightarrow \quad \text{any common } y_T > 0 \text{ is an equilibrium}$$

$\hookrightarrow$ My spending is your income, your spending is my income. The MPC is one: the consumption function lies on the 45° line. Prices need not be "wrong" for effective demand to be indeterminate.

The final morning: optimal relative prices

Now include the morning price choice. Suppress T and write $q_i = p(i)/p$. For given aggregates (p, y), demand and the terminal budget give

$$y(i) = yq_i^{-\gamma} \qquad c(i) = yq_i^{1-\gamma}$$

Write $u'(c) = c^{-\sigma}$. Farmer i maximizes $u(yq_i^{1-\gamma}) - v(yq_i^{-\gamma})$ over q_i. The first-order condition is

$$q_i^{1+\sigma(\gamma-1)} y^{-\sigma} = \frac{\gamma}{\gamma - 1} v'(yq_i^{-\gamma}) \quad \Longrightarrow \quad q_i = f(y) \quad f' > 0 \quad f(y^*) = 1$$

Every farmer chooses the same relative price. The price-index identity then requires

$$q_i = 1 \quad \Longrightarrow \quad y = y^*$$

$\hookrightarrow$ Optimal price setting in the morning selects afternoon output, even though the afternoon subgame itself permits any output level.

What does price flexibility actually change?

Compare two games with the same realized prices and credit balances at the beginning of the final afternoon:

Morning	Afternoon state	Equilibrium output
Prices freely chosen	$p(i) = p \quad m(i) = 0$	$y = y^*$
Prices fixed at that level	$p(i) = p \quad m(i) = 0$	Any $y > 0$

The pricing first-order condition disappears in the fixed-price game. Mathematically, the difference is straightforward.

How does the option to change prices coordinate spending,
when the prices that people actually face are unchanged?

$\hookrightarrow$ The restriction comes from requiring compatibility with earlier optimality. The model has not supplied a process transmitting that restriction to the afternoon: RE$\rightarrow$ Say's law "for free".

The two ingredients of the paradox

1. Subgame multiplicity. Following a history x, several continuations $y \in Y(x)$ are equilibria.
   This comes from indifference in the *ij* game, and strategic complementarities in afternoon spending in the pricing game.

2. Nonexistence for some continuation selections. Choose $y(x) \in Y(x)$ and solve the resulting morning game. Some selections make that game have no equilibrium.

In the *ij* game, any nonzero constant y makes the i agents' payoffs unbounded. In the pricing game, $y < y^*$ makes every farmer want a price below the average: all cannot succeed.

$\hookrightarrow$ Full-game equilibrium rules out these continuations by imposing earlier optimality.
(The strategy sets and the assumption of atomistic agents also matter.)

Temporal locality and hidden states

Temporal locality: history affects behaviour today through something that is still present today — knowledge, beliefs, preferences, opportunities, property rights, etc.

In the pricing game, the same prices and balances permit different predictions depending on how prices were set. To interpret this causally, we need additional state variables.

- Memory that prices were freely chosen, and beliefs about what this implies for other farmers' spending.
- More generally, in Walrasian models: actual price-setting behaviour and rules for handling excess demands.

$\hookrightarrow$ One can postulate beliefs that reproduce equilibrium. The substantive question is whether a plausible belief-formation process would generate them.

Could rational introspection do the work?

The eductive approach: eliminate actions that are never optimal, then eliminate actions that cease to be optimal once others do the same, and continue.

$\hookrightarrow$ Common belief in rationality leads to rationalizable strategies, which need not select the equilibrium.

In the pricing game, pessimistic beliefs $y < y^*$ remain rationalizable:

- Farmer i may optimally set $p_i = 0.9$, expecting an average price of 1.
- She can believe others set 1 because *they* expect an average of 1.1, and so on.

These expectations cannot all be correct. But rationality alone does not make them mutually consistent.

Observing the morning price profile does not force farmers to expect $y = y^*$.

Could learning coordinate expectations?

A repeated, linearized version makes the difficulty transparent. Let x denote log prices, and y the log output deviation from full employment:

$$x_t^i = x_t^e + \kappa y_t^e \qquad y_t^i = y_t^e \qquad \kappa > 0$$

With common beliefs and a decreasing gain $g_t \in (0, 1)$,

$$x_{t+1}^e = x_t^e + g_t(x_t - x_t^e) = x_t^e + g_t \kappa y_t^e$$
$$y_{t+1}^e = y_t^e + g_t(y_t - y_t^e) = y_t^e$$

If initial demand expectations differ from zero, actual spending validates them. Adjusting price expectations does not bring output to full employment.

In Dogra's dynamic pricing game, anticipated-utility learning is also locally unstable under the zero nominal interest-rate peg.

$\hookrightarrow$ Convergence must be established for a specified process and policy regime: price flexibility alone does not establish anything.

1 Motivation

2 Process and equilibrium models

3 Equilibrium and mechanisms

4 Temporal non-locality

5 Immaculate revelation

Learning something that no one knows

Nature draws $\theta \sim \mathcal{N}(0, \sigma^2)$. Agents $i = 1, 2$ minimize $(a_i - \theta)^2$. Neither observes θ; each observes only the other's action.

$$a_1(\theta) = \mathbb{E}[\theta \mid a_2 = a_2(\theta)] \qquad a_2(\theta) = \mathbb{E}[\theta \mid a_1 = a_1(\theta)]$$

Allow the action functions to be measurable with respect to θ. There are at least two RE equilibria:

- Uninformative: $a_1(\theta) = a_2(\theta) = 0$. Each chooses the prior mean.
- Fully revealing: $a_1(\theta) = a_2(\theta) = \theta$. Each learns θ from the other's action.

How did the information get into either action in the first place?

$\hookrightarrow$ It's impossible to describe a causal story that brings this about.

Does Kreps's restriction solve the problem?

Require endogenous variables to be measurable with respect to the information possessed by all agents taken together (Kreps, 1977).

In the preceding example, this rules out $a_1 = a_2 = \theta$. Now add a third agent:

$$\text{Agent 3 observes } \theta \qquad \min_{a_3} a_3^2 \quad \Longrightarrow \quad a_3(\theta) = 0$$

The pooled information now includes θ. Consequently,

$$a_1(\theta) = a_2(\theta) = \theta \qquad a_3(\theta) = 0$$

is compatible with the measurability restriction and each agent's optimality.

$\hookrightarrow$ But agent 3's action reveals nothing. Even observing it cannot transmit her information to agents 1 and 2. The restriction does not describe a route by which information travels.

When immaculate revelation is necessary: setup

Three agents, two goods. Good 2 is the numeraire; good 1 has price p and total endowment 3. Superscripts index goods. Preferences and private information:

Agent	Utility	Observes
1	$\beta \ln x_1^1 + x_1^2$	β
2	$(3-\beta)(1+\varepsilon)\ln x_2^1 + x_2^2$	Neither shock
3	x_3^2	ε

$\beta \in \{1, 2\}$ and $\varepsilon \in \{-\bar{\varepsilon}, \bar{\varepsilon}\}$ are independent, equiprobable; $0 < \bar{\varepsilon} < 1/2$. All agents observe p.

Give agent 3 no good-1 endowment, and agents 1 and 2 ample numeraire wealth. Then agent 3 never trades. Interior good-1 demands are

$$x_1^1 = \frac{\beta}{p} \qquad x_2^1 = \frac{\mathbb{E}[(3-\beta)(1+\varepsilon) \mid p]}{p} \qquad x_3^1 = 0$$

Why prices cannot ignore the hidden information

Suppose there were an RE equilibrium with $p(\beta, \varepsilon)$ independent of ε. Since β has two possible values, there are only two cases:

1. p does not reveal β. Then p is constant. Agent 2 demands $(3 - \mathbb{E}\beta)/p = 3/(2p)$, while agent 1 demands β/p. Market clearing would require

$$\frac{\beta + 3/2}{p} = 3$$

   at the same price for both values of β: impossible.

2. p reveals β. Since $\mathbb{E}\varepsilon = 0$, demands imply

$$\frac{\beta}{p} + \frac{3 - \beta}{p} = 3 \quad \Longrightarrow \quad p = 1$$

   But a constant price cannot reveal β.

$\hookrightarrow$ There is no equilibrium whose price is independent of ε, although only the inactive agent initially knows it.

A fully revealing equilibrium exists

If prices reveal both shocks, market clearing requires

$$\frac{\beta}{p} + \frac{(3-\beta)(1+\varepsilon)}{p} = 3 \quad \Longrightarrow \quad p(\beta, \varepsilon) = 1 + \frac{(3-\beta)\varepsilon}{3}$$

	$\varepsilon = -\bar{\varepsilon}$	$\varepsilon = +\bar{\varepsilon}$
$\beta = 1$	$1 - 2\bar{\varepsilon}/3$	$1 + 2\bar{\varepsilon}/3$
$\beta = 2$	$1 - \bar{\varepsilon}/3$	$1 + \bar{\varepsilon}/3$

The four prices differ, so observing p indeed reveals the state. Optimal demands then validate the price function.

$\hookrightarrow$ Equilibrium requires prices to depend on information that no active trader initially has. This construction reveals it completely. The inactive informed agent satisfies Kreps's restriction, but supplies no causal information channel.

Conclusion

"[...]if we can describe the causal process which causes the assumptions of market clearing and consistent beliefs to hold, we will know what these assumptions mean, we will understand them in a sense that equilibrium models will never be understood. This is exactly why we care about the disequilibrium foundations of equilibrium economics.